

Cloud Attacks

CompTIA PenTest+ (PT0-003)

Domain: 4.0: Attacking Cloud Technologies

Objective: 4.6: Given a scenario, perform cloud-based attacks using the appropriate tools.

Video Overview

This video provides an overview of cloud-based attacks, offering insight into the vulnerabilities and methodologies attackers use in cloud environments. Understanding these attack vectors is important for students preparing for the CompTIA PenTest+ (PT0-003) certification exam, as it directly addresses objectives related to cloud security. For those pursuing careers in IT, this knowledge is relevant for securing cloud infrastructures and performing effective penetration tests.

Key Topics Covered

- **Cloud Vulnerabilities:** Identifying common weaknesses stemming from user misconfigurations rather than cloud provider flaws.
- **Identity and Access Management (IAM) Misconfigurations:** Risks associated with over-permissioning, improper roles, and federation misconfigurations, particularly with Single Sign-On (SSO).
- **Open Object Storage:** Dangers of misconfigured S3 buckets, including unrestricted uploads, public read access, and exposed directory listings.
- **Credential Harvesting:** Methods attackers use to obtain cloud account credentials, such as phishing, malware, social engineering, and brute-force techniques.
- **Account Takeover and Privilege Escalation:** How attackers leverage compromised credentials to gain unauthorized access and expand control within cloud systems (vertical and horizontal escalation).
- **Metadata Service Attacks:** Exploiting metadata to steal temporary tokens and API keys to impersonate services.
- **Resource Exhaustion and Denial of Service (DoS):** Understanding how misconfigurations or user errors can lead to DoS attacks, including direct-to-origin attacks that bypass CDNs.

- **Cloud Malware Injection Attacks:** Malicious images (AMIs) and their potential to compromise systems, including hypervisor targeting.
- **Cloud Deployment and Application Vulnerabilities:** Threats in containerized environments (Kubernetes, AWS ECS), including container escapes and sensitive secret leaks.
- **Side-Channel Attacks:** Recovering residual data from underlying hardware in multi-tenant cloud environments.
- **Cloud Penetration Testing Tools:** Introduction to tools like Scout Suite, Prowler, CloudBrute, Pacu, and Cloud Custodian for mapping, assessing, and exploiting cloud environments.

Student Notes Section